

Calculus of Variation
Assignment 7



Q1 A particle of mass m is moving vertically, under the action of gravity and resistive force numerically equal to c times its velocity $\dot{y}$. Which of the following is/are the form of its Hamilton's principle equation?

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| a) $\delta \int_{t_1}^{t_2} \left(\frac{1}{2} m \dot{y}^2 + mgy \right) dt - \int_{t_1}^{t_2} cy \delta y dt$ | b) $\delta \int_{t_1}^{t_2} \left(\frac{1}{2} m \dot{y}^2 - mgy \right) dt + \int_{t_1}^{t_2} cy \delta y dt$ |
| c) $\delta \int_{t_1}^{t_2} \left(\frac{1}{2} m \dot{y}^2 + mgy \right) dt - \int_{t_1}^{t_2} cy \delta y dt$ | d) $\delta \int_{t_1}^{t_2} \left(\frac{1}{2} m \dot{y}^2 + mgy \right) dt + \int_{t_1}^{t_2} cy \delta y dt$ |
| e) $\delta \int_{t_1}^{t_2} \left(\frac{1}{2} m \dot{y}^2 + mgy \right) dt - \int_{t_1}^{t_2} cy \delta y dt$ | f) $\delta \int_{t_1}^{t_2} \left(\frac{1}{2} m \dot{y}^2 + mgy \right) dt + \int_{t_1}^{t_2} cy \delta y dt$ |

Q2 Three masses are connected in series to a fixed support, by linear springs. Let x_i represent displacement from equilibrium and k_i be the spring constants. Assuming that only the spring forces are present, what would be the lagrangian function(s) of the system?

- a) $L = \left[m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2 + m_3 \dot{x}_3^2 - k_1 x_1^2 - k_2 (x_2 - x_1)^2 - k_3 (x_3 - x_2)^2 \right] + \text{constant}$
- b) $L = \frac{1}{2} \left[m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2 + m_3 \dot{x}_3^2 + k_1 x_1^2 + k_2 (x_2 - x_1)^2 + k_3 (x_3 - x_2)^2 \right]$
- c) $L = \frac{1}{2} \left[m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2 + m_3 \dot{x}_3^2 - k_1 (x_1 - x_3)^2 - k_2 (x_2 - x_1)^2 - k_3 (x_3 - x_2)^2 \right] + \text{constant}$
- d) $L = \frac{1}{2} \left[m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2 + m_3 \dot{x}_3^2 - k_1 x_1^2 - k_2 (x_2 - x_1)^2 - k_3 (x_3 - x_2)^2 \right] + \text{constant}$
- e) $L = \frac{1}{2} \left[m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2 + m_3 \dot{x}_3^2 + k_1 x_1^2 - k_2 (x_2 - x_1)^2 + k_3 (x_3 - x_2)^2 \right] + \text{constant}$
- f) $L = \frac{1}{2} \left[m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2 + m_3 \dot{x}_3^2 - k_1 x_1^2 - k_2 (x_2 - x_1)^2 - k_3 (x_3 - x_2)^2 \right] + \text{constant}$

Q3 A particle moves on the surface $\phi(x, y, z) = 0$ from the point (x_1, y_1, z_1) to the point (x_2, y_2, z_2) in time T . If it moves in such a way that the integral of its kinetic energy over that time is minimum, which of the following equations must the coordinates of the particle satisfy?

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| a) $\frac{\ddot{x}}{\phi_x} = \frac{\ddot{y}}{\phi_y} = \frac{\ddot{z}}{\phi_z}$ | b) $\frac{\dot{x}}{\phi_x} = \frac{\dot{y}}{\phi_y} = \frac{\dot{z}}{\phi_z}$ |
| c) $\frac{x}{\phi_x} = \frac{y}{\phi_y} = \frac{z}{\phi_z}$ | d) $\frac{\ddot{x}}{\phi_x} = \frac{y}{\phi_y} = \frac{z}{\phi_z}$ |
| e) $\frac{x}{\phi_x} = \frac{y}{\phi_y} = \frac{\ddot{z}}{\phi_z}$ | f) $\frac{x}{\phi_x} = \frac{\ddot{y}}{\phi_y} = \frac{z}{\phi_z}$ |